



Authentication

WARNING

If you are trying to authenticate as a user, please refer to this [guide](#) instead.

This guide will go through the authentication mechanisms for service principal accounts.

Snowflake Authentication

TIP

Applicable if you are using the [direct Snowflake connection](#).

If you are using the [Snowflake SDKs](#), the SDKs will implement the authentication mechanism - follow the documentation of the respective SDKs' API.

However, if you are using the [Snowflake REST API](#), you can follow [Snowflake's documentation](#).

DataHive Keypair Authentication

TIP

Applicable if you are using [DataHive Web API](#).

Authenticate your Service Principal Account by generating a JWT (JSON Web Token) signed with your account's private key.

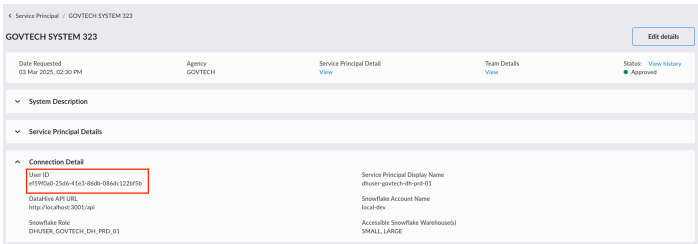
INFO

We have a sample script [here](#) which implements the steps to generate the JWT token.

In the Python script, the method for generating JWT token can be found in `api_service.py`, in `_generate_keypair_token` method.

Alternatively, you can implement your own JWT generator by following the steps below:

1. Generate the `public_key_fingerprint`, a SHA-256 hash, of the public key for the service principal account. Prefix the fingerprint with `SHA256:`.
 - For example: `SHA256:<hash>`
2. Generate a JSON Web Token (JWT) with the following fields in the payload:

Field	Description	Example
iss	Issuer of the JWT. Set it to the following value: <code>GOVTECH-DATAHIVE.<user_id_in_uppercase>.SHA256:<public_key_fingerprint></code> where: 1. user_id_in_uppercase is the ID of the service principal account in uppercase. 2. public_key_fingerprint is the fingerprint that you generated in the step 1. Ensure that the fingerprint is used as is, based on what is generated. The User ID can be found here: 	<code>GOVTECH-DATAHIVE.65180D27-22B5-4635-B461-51D7E3017832.SHA256:<public_key_fingerprint></code>
sub	Subject for the JWT. Set it to the following value: <code>GOVTECH-DATAHIVE.<user_id_in_uppercase></code> where: 1. user_id_in_uppercase is the ID of the service principal account in uppercase.	<code>GOVTECH-DATAHIVE.65180D27-22B5-4635-B461-51D7E3017832</code>

Field	Description	Example
<code>iat</code>	Issue time for the JWT in UTC. Set the value to the current time value as seconds.	<code>1615370644</code>
<code>exp</code>	Expiration time for the JWT in UTC (typically +60 min of <code>iat</code>). You can specify the value as seconds.	<code>1615374184</code>

3. Sign the claims with the Service Principal's private key, which will produce the JWT token required for authentication.

4. In each API request that you send, set the following headers:

- `Authorization: Bearer <JWT>`

DataHive Gateway Key



TIP

Applicable if you are using [DataHive Synchronous API \(Enhanced\)](#).

This is required in addition to the [keypair authentication described above](#).

When a Synchronous API (Enhanced) is provisioned to your Service Principal Account, you will be provided with the Gateway key required to access the Synchronous API.

To authenticate successfully, include the following headers:

- `Authorization: Bearer <JWT>`
- `DATAHIVE-GATEWAY-KEY: <Gateway Key>`